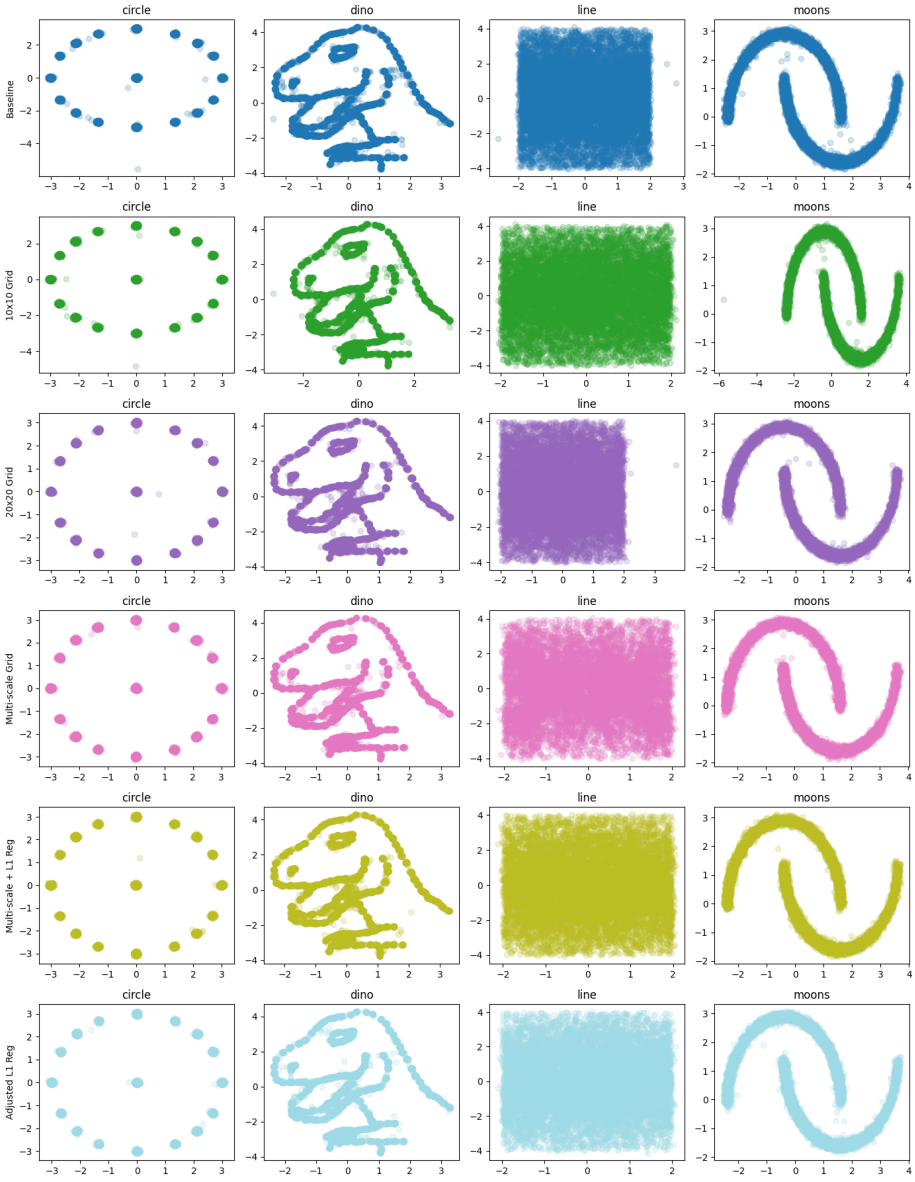


Figure 2: PLEASE FILL IN CAPTION HERE

improvements observed in our metrics, particularly for the more complex shapes like the dino and moons datasets.

As shown in Table 1, our proposed models incur increased training times compared to the baseline DDPM. The multi-scale grid approach with L1 regularization takes approximately 79% longer to train. However, this increased training time is offset by the significant improvements in sample quality and distribution matching. Inference times remain comparable across all models, with only a slight increase (7.9% for our full model) relative to the baseline.

Figure 3 shows the training loss over time for each dataset across all model configurations.

The training loss curves demonstrate consistent convergence across all datasets, with our multi-scale grid approaches showing faster initial decreases in loss compared to the baseline DDPM. The L1-regularized version exhibits slightly higher final training loss, which aligns with our observations of

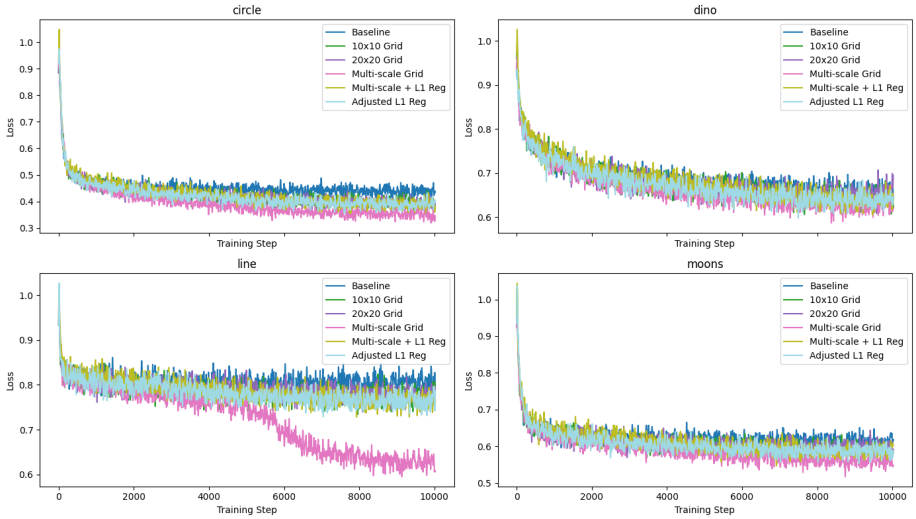


Figure 3: Training loss over time for each dataset (circle, dino, line, and moons) across all runs.

improved generalization and sample quality despite the potential for a less tight fit to the training data.

Our ablation studies reveal the individual contributions of the multi-scale approach and L1 regularization:

1. Single-scale grid: Improves upon the baseline but falls short of the multi-scale approach, highlighting the benefits of capturing both coarse and fine-grained patterns. 2. Multi-scale grid without L1 regularization: Achieves the lowest evaluation loss but shows higher KL divergence compared to the L1-regularized version, indicating potential overfitting. 3. Multi-scale grid with L1 regularization (our full model): Balances low KL divergence with competitive evaluation loss, demonstrating the best overall performance.

Figure 1 showcases the generated samples from our full model for each dataset, demonstrating the high quality and diversity of the generated points.

Despite the overall improvements, our method has some limitations: 1. Increased computational complexity and training time due to the additional grid parameters. 2. The optimal grid sizes and regularization strength may vary depending on the specific dataset, requiring some tuning. 3. The effectiveness of the method on higher-dimensional (e.g., 3D or 4D) datasets remains to be explored.

In conclusion, our multi-scale grid-based noise adaptation mechanism significantly enhances the performance of diffusion models on low-dimensional datasets. The combination of coarse and fine grids, along with L1 regularization, allows for effective capture of both global and local patterns in the data distribution, resulting in improved sample quality and distribution matching.

7 Conclusions and Future Work

In this paper, we introduced a novel multi-scale grid-based noise adaptation mechanism for enhancing the performance of diffusion models on low-dimensional datasets. Our approach addresses the unique challenges posed by low-dimensional data by employing a combination of coarse (5×5) and fine (20×20) grids to dynamically adjust noise levels during the diffusion process. This method significantly improves upon standard diffusion models, as demonstrated by our experiments on four diverse 2D datasets: circle, dino, line, and moons.

Key contributions and findings of our work include:

1. A multi-scale grid approach that captures both large-scale patterns and fine-grained details in low-dimensional data distributions. 2. Significant reductions in KL divergence, with improvements

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of up to 16.83. Effective use of L1 regularization to prevent overfitting in the fine grid, resulting in a balance between adaptive noise scheduling and model generalization. 4. Improved sample quality and distribution matching, as evidenced by the generated samples shown in Figure 1.

Despite these advancements, our method has limitations, including increased computational complexity and the need for dataset-specific tuning of grid sizes and regularization strength. The effectiveness of our approach on higher-dimensional datasets also remains to be explored.

Future work directions include:

1. Extending the method to higher-dimensional datasets (3D, 4D, etc.) to broaden its applicability. 2. Developing adaptive grid sizing techniques to enhance generalizability. 3. Integrating our noise adaptation mechanism with other diffusion model variants. 4. Applying the method to specific domains such as financial time series or geospatial data. 5. Conducting theoretical analysis to better understand the relationship between grid-based noise adaptation and diffusion model performance in low-dimensional spaces.

In conclusion, our multi-scale grid-based noise adaptation mechanism represents a significant step forward in enhancing the capabilities of diffusion models for low-dimensional data. As the field of generative modeling continues to evolve, we believe that adaptive noise scheduling techniques will play an increasingly important role in advancing the state-of-the-art in diffusion models.

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